

RESEARCH ARTICLE | Physical Activity and Inactivity

Rest interval duration does not influence adaptations in acid/base transport proteins following 10 wk of sprint-interval training in active women

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Submitted 27 October 2016; accepted in final form 30 January 2017

McGinley C, Bishop DJ. Rest interval duration does not influence adaptations in acid/base transport proteins following 10 wk of sprint-interval training in active women. *Am J Physiol Regul Integr Comp Physiol* 312: R702–R717, 2017. First published February 1, 2017; doi:10.1152/ajpregu.00459.2016.—The removal of protons (H^+) produced during intense exercise is important for skeletal muscle function, yet it remains unclear how best to structure exercise training to improve muscle pH regulation. We investigated whether 4 wk of work-matched sprint-interval training (SIT), performed 3 days/wk, with either 1 (*Rest-1*; $n = 7$) or 5 (*Rest-5*; $n = 7$) min of rest between sprints, influenced adaptations in acid/base transport protein content, nonbicarbonate muscle buffer capacity ($\beta_{m, \text{in vitro}}$), and exercise capacity in active women. Following 1 wk of posttesting, comprising a biopsy, a repeated-sprint ability (RSA) test, and a graded-exercise test, maintenance of adaptations was then studied by reducing SIT volume to 1 day/wk for a further 5 wk. After 4 wk of SIT, there was increased protein abundance of monocarboxylate transporter (MCT)-1, sodium/hydrogen exchanger (NHE)-1, and carbonic anhydrase (CA) XIV for both groups, but rest interval duration did not influence the adaptive response. In contrast, greater improvements in total work performed during the RSA test after 4 wk of SIT were evident for *Rest-5* compared with *Rest-1* (effect size: 0.51; 90% confidence limits ± 0.37), whereas both groups had similarly modest improvements in $\dot{V}O_{2\text{peak}}$. When training volume was reduced to 1 day/wk, enhanced acid/base transport protein abundance was maintained, although NHE1 content increased further for *Rest-5* only. Finally, our data support intracellular lactate as a signaling molecule for inducing MCT1 expression, but neither lactate nor H^+ accumulation appears to be important signaling factors in MCT4 regulation.

lactate transport; intracellular buffering; repeated-sprint ability; de-training

HIGH-INTENSITY EXERCISE demands a large nonmitochondrial ATP turnover and results in accumulation of protons (H^+) in skeletal muscle, which have been implicated in excitation-contraction coupling inhibition (69), impairment of cross-bridge cycling (27), and inactivation of key glycolytic enzymes (74). Mitigation of the exercise-induced reduction in intracellular pH (pH_i) entails uptake of H^+ by physicochemical buffers, such as histidyl-imidazole residues and inorganic phosphate (P_i), and buffering of H^+ by metabolic reactions in parallel with energy production (48). Acid/base transport across the sarcolemma by specific proteins provides a means to

remove H^+ from the myofiber or to introduce bicarbonate or equivalents (CO_3^{2-} , $NaCO_3^-$) (50, 77). Given the importance of pH_i regulation for muscle function and exercise capacity, much research has focused on the effects of exercise training on acid/base transport proteins and muscle buffer capacity (β_m). Most of these studies have employed different modalities of high-intensity interval training [HIIT; repeated exercise bouts at intensities approaching 100% maximal oxygen uptake ($\dot{V}O_{2\text{max}}$)] or sprint interval training [SIT; repeated exercise bouts at intensities $\geq 100\% \dot{V}O_{2\text{max}}$] (84). However, equivocal findings to date indicate the stimuli required for upregulation of the individual components of pH_i regulation remain to be determined.

The acid/base transport proteins can be categorized as either lactate (La^-) coupled or nonlactate coupled. The former comprises the monocarboxylate transport proteins (MCT1 and MCT4) and their chaperone protein basigin. The latter consists of the sodium/hydrogen exchanger (NHE) proteins, with the skeletal muscle isoform being NHE1. H^+ -equivalent transport is provided by the sodium-coupled bicarbonate transporter (NCBT) family of proteins, of which, isoform-specific protein evidence has been shown for only the electrogenic sodium/bicarbonate cotransporter (NBCe)-1 (63). In addition, a functional, and perhaps physical, interaction has been demonstrated in vitro between each of these acid/base transport proteins and the skeletal muscle carbonic anhydrase (CA) isozymes (6). While there has been much investigation of the response to training of the MCTs and NHE1, although little of these other proteins, the signaling networks responsible for regulation of any of the acid/base transport proteins remain poorly understood compared with, for example, the well-described protein kinase B/mammalian target of rapamycin and AMP-activated protein kinase/proliferator-activated receptor- γ coactivator-1 α pathways for resistance and endurance exercise, respectively (e.g., see Ref. 29).

Notwithstanding this limited picture of transcriptional and translational events for the acid/base transporters, there must be an initial signal(s) activated by exercise that initiates a phenotypic change (65). One such signal previously proposed for different acid/base transporters is the accumulation, or sustained production, of intracellular La^-/H^+ (38, 51, 53, 80). Through manipulating the rest period between high-intensity work intervals, it should be possible to design contrasting exercise protocols that result in either high absolute La^-/H^+ levels or sustained production of La^-/H^+ . For example, muscle pH has been shown to decrease even further in the first 90

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s after a single 30-s sprint (17); therefore, exercise comprising high-intensity work intervals, interspersed with rest periods <90 s, should result in large peaks in La^-/H^+ . However, given insufficient recovery of pH_i between intervals may result in reduced glycogen phosphorylase activity during subsequent intervals (74), sustained production of La^-/H^+ may require longer rest between work intervals. With muscle pH recovering by ~50% within 4–6 min after single or multiple sprints (8, 17, 64), similar-duration rest periods may maximize anaerobic glycolysis for sequential intervals (74). Ultimately, by manipulating rest interval duration, while matching intensity and total work, it may be possible to test whether different exercise-induced changes in intracellular La^-/H^+ subsequently influence upregulation of specific acid/base transport proteins.

In addition to the uncertainty as how best to elicit improvements in pH_i -regulatory components, it is also not clear how detraining might be mitigated. Adaptations of acid/base transport proteins to training have been shown to be transient following cessation of HIIT. In a previous study (63), we found almost total reversal of adaptations in all of the acid/base transporters 6 wk after completing 4 wk of HIIT. Reductions in MCT1 and MCT4 abundance have similarly been reported 6 wk after stopping SIT (18). It is known that maintenance of some training-induced adaptations can be achieved, or dysregulation mitigated, by reducing the volume of training while maintaining the intensity (41, 42, 46, 47, 67). Although it is possible that fractional degradation of some or all the acid/base transport proteins can be mitigated by a reduced volume of high-intensity training, this hypothesis has not been tested.

The present study therefore compared two different recovery durations between 30-s sprints during work-matched SIT. We hypothesized that, through altering the duration of the rest interval, subsequent variations in intracellular La^-/H^+ would influence the adaptive responses of the acid/base transport proteins. In addition, we postulated that improvements in βm and exercise performance would reflect potential differences in substrate usage. Finally, we proposed that performing SIT 1 day/wk for 5 wk, following a 4-wk period of training 3 days/wk, would be sufficient to maintain any potential adap-

tations in acid/base transport proteins, βm , and repeated-sprint ability.

MATERIALS AND METHODS

Participants

Twenty-one recreationally active women gave written informed consent to participate in this study. Six participants withdrew during the familiarization period because of either unrelated illness/injury, or a lack of time, and one participant withdrew after the first SIT session (SIT1) because of an unwillingness to undergo further biopsies. Fourteen participants completed the entire study [age: 24 (6) yr; height: 164.6 (3.8) cm; mass: 62.9 (7.8) kg; $\text{VO}_{2\text{peak}}$: 39.3 (5.3) $\text{ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$; mean (SD)]. Before the training intervention began, weekly training load was 1,005 (593) arbitrary units [mean (SD)], calculated from session rating of perceived exertion and training duration (28). All procedures were approved by the Victoria University Human Research Ethics Committee.

Experimental Design

The study employed a two-group parallel design (Fig. 1). There were two work-matched SIT groups, differing only in the duration of the rest component of the duty cycle, either 1-min rest (*Rest-1*; $n = 7$) or 5-min rest (*Rest-5*; $n = 7$). Following familiarization trials and pretraining tests, participants were ranked on total work performed during a repeated-sprint ability (RSA) test and coded consecutively with either an A or a B in ABBA format. With the use of random generation of binary numbers, type of training was then randomly allocated to *group A* (*Rest-5*) and *group B* (*Rest-1*).

The study consisted of two training phases: participants trained 3 days/wk for the first 4 wk, and, following 1 wk of posttesting, they then trained 1 day/wk for 5 wk. Before (Pre) and after 4 (+4 wk) and 10 (+10 wk) wk of training, the following experimental trials were conducted at least 2 days apart: 1) an RSA test, 2) a graded-exercise test, followed 5 min later by a square wave peak oxygen uptake ($\text{VO}_{2\text{peak}}$) test (Pre and +4 wk only), and 3) a resting muscle biopsy. The resting muscle biopsy Pre was conducted before the first training session, with a second biopsy (Post) taken immediately following exercise (detailed in SIT1 subsection).

Participants were familiarized with all exercise tests before undertaking experimental trials. They performed two familiarization trials of the RSA test on separate days, and on another day they performed

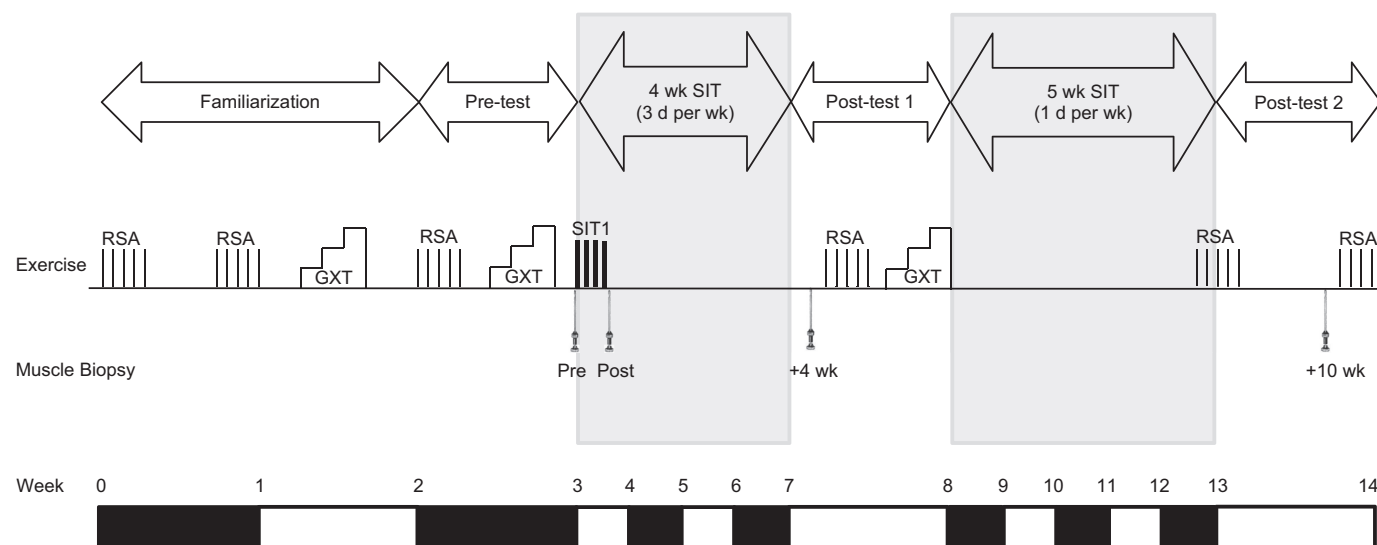


Fig. 1. Experimental design. GXT, graded-exercise test; RSA, repeated-sprint ability test; SIT1, sprint-interval training session 1.

familiarization trials of the graded-exercise test and $\dot{V}O_{2\text{peak}}$ test. In the week preceding the +10-wk trial, participants performed a re-familiarization of the RSA test.

Sprint-Interval Training

The training intervention consisted of supervised work-matched SIT performed on an electromagnetically braked cycle ergometer (Velotron; Racer-Mate, Seattle, WA). Participants performed a standardized 5-min steady-state warm-up at 60 W, followed by 4 × 30-s sprints in the first session, increasing to a maximum of 10 × 30-s sprints in the 10th session (see Table 1 for training program). All sprints were interspersed with either 1 min (*Rest-1*) or 5 min (*Rest-5*) of passive recovery. Training intensity was initially set at 200% of the power at the pretraining lactate threshold (LT), increasing to 305% of the pretraining LT by week 4, which was equivalent to between 121 (14)% and 186 (17)% of baseline peak aerobic power (W_{peak}) during the first 4 wk. The 13th session training intensity was adjusted to 300% of the +4 wk LT or, where necessary, to avoid a reduction in intensity, adjusted to up to 350% of the LT. This was equivalent to between 190 (13)% and 204 (14)% of the +4 wk W_{peak} during weeks 6 to 10.

Experimental Trials

All exercise and/or biopsy trials were performed in the morning (0600–1100) following an overnight fast, with participants recording and subsequently replicating a food diary for the 24 h before each exercise trial. They were also instructed not to drink caffeine on any trial day, and to abstain from drinking alcohol or exercising in the preceding 24 h, with compliance verified by pretrial questionnaires. Participants were asked to maintain their normal level of activity throughout the study and completed a bespoke web-based daily training diary to monitor compliance.

SIT1

To characterize both training protocols, changes in blood and muscle pH and La^- concentration ($[\text{La}^-]$), and βm were measured in response to the first SIT session. The exercise bout consisted of 4 × 30-s sprints at 200% LT, with either 1 or 5 min of rest between sprints, as per group allocation. Muscle biopsies were taken at rest before exercise (Pre) and immediately following the final sprint (Post). Venous blood samples were taken at rest (before the muscle

biopsy) and 2, 3, 5, 7, and 10 min after the final sprint. A 22-gauge intravenous cannula was inserted in an antecubital vein. Before each sample, the cannula was flushed with a small volume of sterile saline, and then ~1 ml of blood was drawn in a dry syringe and discarded. Samples were taken with the participant recumbent on a plinth. Approximately 3 ml of blood were drawn in a heparinized syringe (Rapidlyte; Siemens Healthcare, Melbourne, VIC, Australia). Air was immediately removed from the sample by ejecting blood in the cap and carefully tapping the syringe. The syringe was then gently rolled to mix the sample thoroughly. Unless the sample was ready to be analyzed without delay, the syringe was placed in an ice slurry to minimize changes in pH (23) and subsequently analyzed for blood pH (Rapidpoint 405; Siemens Healthcare). Before this, ~0.5 ml of blood was dispensed in a microtube and immediately analyzed for lactate (2300 STAT Plus; YSI, Yellow Springs, OH), or kept in an ice slurry for imminent analysis.

Needle Muscle Biopsies

Muscle biopsies were taken from the belly of the vastus lateralis, approximately halfway between the knee and hip, using the needle biopsy technique modified with suction (25). Subsequent samples were taken ~1 cm from a previous biopsy site. An incision was made under local anesthesia (1% Xylocaine), and a muscle sample was taken using a Bergström needle (9). Samples were blotted on filter paper to remove blood, before being immediately snap-frozen in liquid nitrogen, and then stored at -80°C until subsequent analyses. Muscle samples were taken from the dominant leg before (Pre) and immediately after (Post) the first training session, 3 days after the 12th training session (+4 wk), and 3 days after the 17th training session (+10 wk).

Repeated-Sprint Ability Test

An isokinetic RSA test was conducted on a separate day before training commenced (Pre), and after 4 (+4 wk) and 10 (+10 wk) wk of SIT. The test was performed at 110 rpm while seated on an electromagnetically braked cycle ergometer (Excalibur Sport, Lode, Groningen, The Netherlands). It followed the same protocol to that previously reported (63), comprising 5 × 6-s maximal sprints, separated by 24 s of rest. During the recovery period, participants remained stationary. Five seconds before the start of each sprint, participants assumed the ready position (crank of the dominant leg at

Table 1. *Sprint-interval training programs performed by the 1- and 5-min rest training groups*

Week	Session No.	No. of Intervals	<i>Rest-1</i> Group		<i>Rest-5</i> Group	
			Relative intensity, %	External power, W	Relative intensity, %	External power, W
1	1	4	200 (0)	221 (33)	200 (0)	223 (45)
	2	5	236 (25)	262 (54)	236 (25)	259 (32)
	3	6	250 (1)	277 (41)	252 (5)	280 (55)
2	4	7	252 (5)	278 (39)	257 (10)	285 (51)
	5	6	277 (5)	306 (43)	273 (12)	303 (52)
	6	8	279 (10)	308 (41)	275 (13)	304 (49)
3	7	9	291 (5)	321 (45)	289 (13)	319 (52)
	8	7	305 (6)	337 (47)	301 (13)	333 (57)
	9	9	305 (6)	337 (47)	305 (17)	336 (53)
4	10	10	305 (6)	337 (47)	307 (22)	339 (52)
	11	7	305 (6)	337 (47)	308 (24)	343 (56)
	12	6	305 (6)	337 (47)	307 (22)	339 (52)
6	13	6	300 (1)	359 (42)	313 (22)	347 (62)
7	14	8	309 (4)	369 (44)	323 (22)	358 (64)
8	15	7	319 (4)	381 (45)	333 (22)	369 (66)
9	16	9	323 (4)	387 (46)	338 (22)	375 (68)
10	17	6	323 (4)	387 (46)	338 (22)	375 (68)

Data are means (SD). Exercise performance tests and a muscle biopsy were performed in week 5. *Rest-1*, 1-min rest training group; *Rest-5*, 5-min rest training group. Relative intensity for weeks 1 to 4 was calculated as a percentage of the lactate threshold (LT) determined during the pretraining graded-exercise test. Relative intensity for weeks 6 to 10 was calculated as a percentage of the LT determined during the +4 wk graded-exercise test.

an angle of 45°). Initially, participants performed a 5-min steady-state warm-up at 60 W followed by 2 × 3-s practice sprints at 90% of perceived maximal effort, separated by 24 s of rest. A criterion 6-s maximal sprint was then performed 90 s later, followed by 5 min of rest before the RSA test. To minimize pacing, participants were required to achieve ≥90% of their criterion score in the first sprint of the 5 × 6-s test. If this criterion was not achieved, participants were required to rest for a further 5 min before restarting the test (one *Rest-1* participant's +10 data were later excluded from the analyses for failing to meet this criterion). Before the first practice sprint, the single 6-s sprint, and the first sprint of the 5 × 6-s test, participants were required to accelerate to 110 revolutions/min and stop pedaling instantly, followed 24 s later by the requisite sprint. This ensured similar velocity of the flywheel before each sprint. Consistent verbal encouragement was given throughout. Total work and work decrement were calculated from the raw data as follows (14):

$$\text{work decrement (\%)} = \left(\frac{\text{total work} - \text{ideal work}}{\text{ideal work}} \times 100 \right)$$

where total work is actual work performed in all five sprints, and ideal work is work performed in *sprint number 1* times five.

Graded-Exercise Test

Graded-exercise tests were performed pretraining and at +4 wk to determine the LT from venous blood samples (taken from an antecubital vein, as described above), and W_{peak} was also recorded. Blood samples were taken at rest and at the end of every stage during the graded-exercise test. Samples were drawn in a 3-ml dry syringe and separated into aliquots in a microtube for instant analysis of lactate (2300 STAT Plus; YSI). The tests were performed seated on the same ergometer used for the RSA test, employing an intermittent protocol, with 4-min exercise stages and 30-s rest stages. Beginning at 50 W, the power was subsequently increased by 25 W every 4.5 min, and consistent verbal encouragement was provided for the latter stages. Participants were required to maintain a set cadence that was self-selected during the familiarization trial and repeated for each of their subsequent graded-exercise tests. The test was terminated either volitionally by the participant or by the assessors when the participant could no longer maintain the required cadence (± 10 revolutions/min), despite strong verbal encouragement. The LT was identified as the power at which venous blood lactate increased by 1 mM above baseline and was calculated using Lactate-E version 2.0 software (72). W_{peak} was calculated as previously reported (57):

$$W_{\text{peak}} = W_{\text{final}} + \left(\frac{t}{240} \times 25 \right)$$

where W_{final} is the power output of the last completed stage, and t is the time in seconds of any final incomplete stage.

Peak Oxygen Uptake Test

Immediately after the graded-exercise test, participants cycled for 5 min at 20 W, followed by a square-wave $\dot{V}O_{2\text{peak}}$ test at 105% of W_{peak} achieved during the graded-exercise test. Participants were instructed to accelerate to 90–100 revolutions/min at the commencement of a 5-s countdown and to maintain a high, but not fixed, cadence until volitional fatigue. Consistent verbal encouragement was provided throughout. Expired gases were analyzed every 15 s using a custom-made metabolic cart. A two-point calibration of the gas analyzers (S-31A/II and CD-3A analyzers; Ametek, Berwyn, PA) was performed before each test using one certified gravimetric gas (16.1% O_2 , 4.17% CO_2 ; BOC Gases, Chatswood, New South Wales, Australia) and ambient air. Ventilation was recorded every 15 s. The ventilometer (KL Engineering, Sunnyvale, CA) was calibrated at the start of each day using a 3-liter syringe (MedGraphics, St. Paul, MN).

$\dot{V}O_{2\text{peak}}$ was calculated as the mean of the two highest consecutive 15-s values. Exercise duration was 145 (30) s [mean (SD)].

Muscle Buffer Capacity

Nonbicarbonate muscle buffer capacity ($\beta_{\text{min vitro}}$) was measured in duplicate on 2–3 mg dry mass of freeze-dried muscle samples using the titration technique (60). Samples were dissected free of visible blood and connective tissue and then homogenized on ice for 3 × 30 s in a 10 mM solution of the glycolytic inhibitor NaF (0.1 ml of NaF/3 mg dry mass). pH measurements were performed at 37°C with a glass microelectrode (MI-410; Microelectrodes, Bedford, NH) connected to a pH meter (Laboratory 850; Schott Instruments, Mainz, Germany). After the initial pH measurement, homogenates were adjusted, if necessary, to pH 7.1–7.2 with 0.02 M NaOH and then titrated to pH 6.1–6.2 with the serial addition of 2 μ l of 0.01 M HCl. From linear regression, the number of moles of H^+ required to change pH from 7.1 to 6.5 was interpolated and then expressed per whole pH unit and relative to sample mass. The typical error (within-sample SD) for repeat titrations was 10.3 mmol H^+ ·kg dry mass⁻¹·pH⁻¹, equivalent to a coefficient of variation (CV) of 7.7% (16).

To determine the potential short-term effects of the different SIT protocols on muscle buffer capacity, nonprotein $\beta_{\text{min vitro}}$ was also assayed on the pre- and post-SIT1 samples using the titration technique (11). Briefly, the homogenate was deproteinized with the addition of 0.03% 5-sulfosalicylic acid hydrate solution (100% wt/vol in 10 mM NaF) and centrifuged for 10 min at 1,000 g. Following a similar procedure to above, the pH of the supernatant was adjusted to 7.1–7.2 with the minimum volume of strong concentration NaOH and then titrated to pH 6.1–6.2 with the serial addition of 2- μ l aliquots of 5 mM HCl. Nonprotein $\beta_{\text{min vitro}}$ was calculated as above, with the protein $\beta_{\text{min vitro}}$ estimated from the difference between whole muscle and nonprotein $\beta_{\text{min vitro}}$. The typical error for repeat titrations was 6.7 mmol H^+ ·kg dry mass⁻¹·pH⁻¹, equivalent to a CV of 10.4% (16).

Muscle Lactate Assay

Approximately 4 mg of freeze-dried pre- and post-SIT1 muscle samples were assayed enzymatically for lactate by spectrophotometric analysis (xMark microplate absorbance spectrophotometer; Bio-Rad Laboratories, Hercules, CA) using a modification of a previously described assay (75). Samples were dissected free of visible blood and connective tissue and then homogenized in 0.6 M perchloric acid using a TissueLyser II (Qiagen, Valencia, CA). Homogenates were then centrifuged at 10,000 g for 10 min at 4°C, and the deproteinized supernatant was removed. KOH (1.2 M) and HEPES (0.125 M) were added, and the supernatant was adjusted to pH 9.0, with 3.0 M KCl then added. After further centrifugation at 10,000 g for 10 min at 4°C, the perchlorate salts were discarded, and the supernatant was stored at -80°C until analysis. Lactate content was measured in triplicate against nine dilutions of a sodium lactate standard (L7022; Sigma-Aldrich, St. Louis, MO). The typical error was 1.5 mmol H^+ /kg dry mass, equivalent to a CV of 2.4% (15).

Quantitative Western Blotting

Muscle homogenate preparation. Approximately 30 mg of frozen muscle tissue were homogenized (Kontes Pellet Pestle, Kimble Chase, NJ) in a 1:20 dilution of ice-cold RIPA buffer (pH 7.4) containing 0.15 M NaCl, 1% Triton X-100, 0.5% sodium deoxycholate, 0.05 M Tris, 0.1% SDS, 1 mM EDTA, 1 × protease/phosphatase inhibitor cocktail (5872; Cell Signaling Technology, Danvers, MA), and 1 mM PMSF. NBCE1 was assayed in muscle tissue homogenized in a HEPES-sucrose buffer (pH 7.4) containing 210 mM sucrose, 30 mM HEPES, 40 mM NaCl, 2 mM EGTA, 5 mM EDTA, 1% protease inhibitor cocktail (P8340; Sigma-Aldrich), 1% phosphatase inhibitor cocktail (P5726; Sigma-Aldrich), and 1 mM PMSF. Homogenates were rotated end-over-end for 60 min at 4°C and centrifuged two

times at 15,000 g for 10 min at 4°C. The supernatants were collected, and the pellets were discarded.

Protein assay. Protein content of muscle homogenate was measured in triplicate using a Bradford assay (Bio-Rad protein assay dye reagent concentrate) against serial dilutions of bovine serum albumin standards (A9647; Sigma-Aldrich).

Immunoblotting. Immunoblotting for each antibody has been previously described in detail, including validation steps taken for specific antibodies (63). Briefly, homogenate was diluted in Laemmli buffer, and equal amounts of total protein (1–25 µg) were loaded in different wells on 10 or 12% Tris-glycine-HCl SDS-PAGE gels or in Criterion 4–12% Bis-Tris SDS-PAGE gels (Bio-Rad) for NBCe1 and CAIV. All samples for a participant were loaded on the same gel. Five or six different dilutions of a mixed-homogenate internal standard were also loaded on every gel, and a calibration curve was plotted of density against protein amount. From the subsequent linear regression equation, protein abundance was calculated from the measured band intensity for each sample on the gel (68). Coomassie blue (Phastgel Blue R-350; GE Healthcare, Rydalmere, New South Wales, Australia) staining of total protein was used as a loading control.

Gel electrophoresis ran for 60–80 min at 100 or 140 V, and proteins were then wet transferred at 100 V to a 0.2-µm polyvinylidene fluoride membrane. Membranes were blocked for 60 min at room temperature in 5% nonfat dry milk diluted in Tris-buffered saline with 0.1% Tween 20 (TBST). After TBST wash steps were completed, membranes were incubated either overnight or for 2 h with the appropriate primary antibody (see Table 2 for conditions). After further TBST washes, the membranes were incubated for 90 min at room temperature in the appropriate secondary antibody. After further washes, membranes were incubated in chemiluminescent solution for 2 min, and images were taken with a VersaDoc Imaging System (Bio-Rad) fitted with a CCD camera (Bio-Rad). Densitometry was performed with Image Laboratory 5.0 software (Bio-Rad) using the volume calculation, and background correction was applied individually to each lane using a rolling-ball algorithm. Images of example blots are displayed in their entirety, except where membranes were cut as described. CVs of replicate gels for individual proteins are presented in their respective figure captions.

A Pre sample for one *Rest-5* participant was blood contaminated, necessitating exclusion of her Western blot data for MCT1, NHE1, NBCe1, and CAII. In addition, there was insufficient muscle biopsied to allow analyses of all proteins for another *Rest-5* participant's +10 wk sample. Consequently, there are no +10 wk data for CAIV, CAXIV, and NBCe1 for this participant. Finally, in analyzing NBCe1 at +10 wk for one *Rest-1* participant, the signal for this band was below the threshold of detection.

Statistical Analyses

Measures of centrality and dispersion are mean (SD) unless otherwise stated. To reduce bias from nonuniformity of error, data were logarithmically transformed where heteroscedasticity was present (70), such as for Western blot data. For these data, geometric mean and SD are reported (geometric mean $\times / \div$ SD). Data were analyzed using linear mixed models with, for example, “time” (repeated measure), “group,” and “group $\times$ time” as fixed factors and “subject” and “intercept” as random factors. First-order autoregressive covariance structures were used for all models, and model fit was assessed by -2 log likelihood (26). Effect sizes (ES) were assessed using Cohen's d , with ES thresholds defined as trivial <0.2 , small <0.6 , moderate <1.2 , large <2.0 , very large <4.0 , and extremely large ≥ 4.0 (45). Uncertainty of effects is expressed as 90% confidence limits (90% CL) and P values, with the latter presented as precise values unless $P < 0.001$ (19). Effects were not considered meaningful if there was $<75\%$ probability of being substantially positive relative to the smallest worthwhile change (ES = 0.2) and were deemed unclear if there was a $>5\%$ probability of also being substantially negative (44, 45). If no between-group differences were found, within-group effects are reported for pooled group data. Linear mixed models were analyzed using IBM SPSS Statistics version 21 (IBM, Somers, NY), and effect sizes and CL were calculated using custom Excel spreadsheets (44). Total work performed by the two groups during the training intervention was compared by an independent-sample unequal-variance t -test and by effect size and CL calculated from a spreadsheet for comparison of two independent groups (43).

RESULTS

Training Data

Both training groups performed similar volumes of total work throughout the training intervention, 1.19 (0.15) and 1.15 (0.23) MJ for *Rest-1* and *Rest-5*, respectively ($t_{12} = 0.37$, $P = 0.72$, ES: 0.18; 90% CL ± 0.90). There was excellent training compliance, with only a single training session being missed by one participant.

Physiological Responses to SIT1

Short-term physiological responses to SIT1 are shown in Figs. 2 and 3. Following SIT1, muscle $[La^-]$ increased to 84.2 (20.3) and 61.8 (31.9) mmol/kg dry mass for *Rest-1* and *Rest-5*, respectively [time main effect: $F(1,15) = 209.26$, $P < 0.001$,

Table 2. Details of primary and secondary antibodies used for Western blotting

Antibody	Supplier	Catalog No./Lot No.	Primary Concentration	Secondary Concentration
<i>Primary</i>				
Rabbit polyclonal anti-MCT1	Merck Millipore	AB3540P/2136555	1:1,000	1:15,000
Rabbit polyclonal anti-MCT4	Merck Millipore	AB3316P/2397059	1:1,000	1:20,000
Mouse monoclonal anti-basigin	Santa Cruz	sc-21746/K1913	1:200	1:7,500
Mouse monoclonal anti-NHE1	Merck Millipore	MAB3140/2283852	1:500	1:7,500
Rabbit polyclonal anti-NBCe1	Cell Signaling	11867/0001	1:500	1:5,000
Rabbit polyclonal anti-CAII	Santa Cruz	sc-25596/F0611	1:1,250	1:30,000
Mouse monoclonal anti-CAIII	Abnova	H00000761-M02/12243-S1	1:2,500	1:40,000
Mouse polyclonal anti-CAIV	Abnova	H00000762-B02P/08325 WULz	1:500	1:10,000
Mouse polyclonal anti-CAXIV	Abnova	H00023632-B01P/08358 WULz	1:750	1:10,000
<i>Secondary</i>				
Goat anti-mouse IgG	PerkinElmer	NEF822001EA		
Goat anti-rabbit IgG	PerkinElmer	NEF812001EA		

MCT, monocarboxylate transporter; NHE1, sodium/hydrogen exchanger-1; NBCe1, electrogenic sodium/bicarbonate cotransporter-1; CA, carbonic anhydrase.

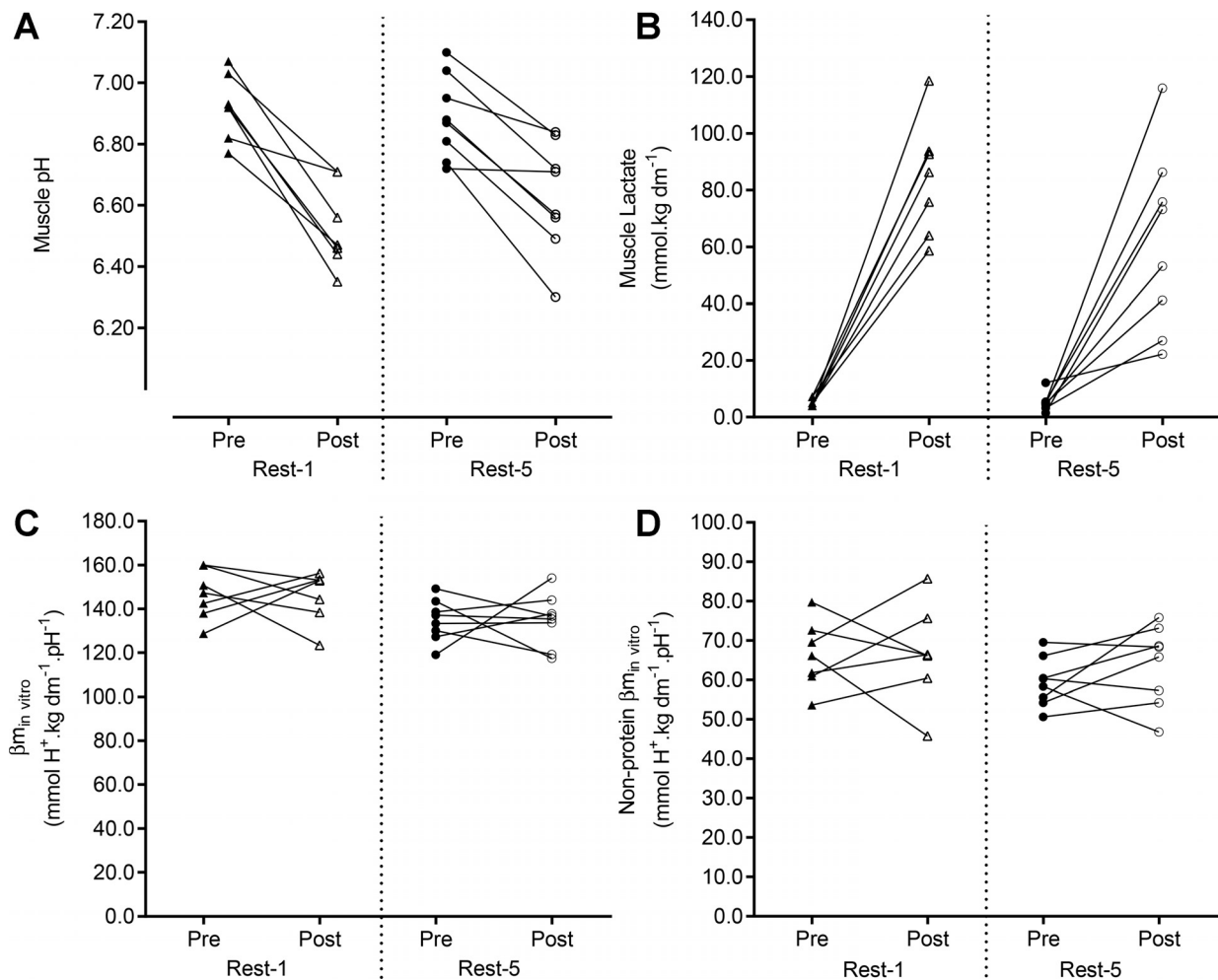


Fig. 2. Muscle pH (A), muscle lactate (B), total nonbicarbonate muscle buffer capacity ($\beta_{m_{in\ vitro}}$, C), and nonprotein $\beta_{m_{in\ vitro}}$ (D) before (Pre) and immediately after (Post) a single bout of sprint-interval training (SIT1) for *Rest-1* (triangles, $n = 7$) and *Rest-5* (circles, $n = 8$; including one participant who dropped out after SIT1). Individual data points are plotted. CV of duplicate samples: pH (0.6%), lactate (3.8%), total $\beta_{m_{in\ vitro}}$ (6%), and nonprotein $\beta_{m_{in\ vitro}}$ (10.4%).

pooled ES: 5.49, 90% CL $\times / \div 0.72$), but there was no clear difference between groups [group $\times$ time interaction: $F(1,15) = 0.86$, $P = 0.37$, ES: 0.70, $\times / \div 1.42$]. Assuming a muscle water content of 3.3 l/kg dry mass (48), this equates to 25.5 (6.2) and 18.7 (9.7) mM for *Rest-1* and *Rest-5*, respectively. Muscle pH decreased more for *Rest-1*, 6.92 (0.11) to 6.53 (0.14), than *Rest-5*, 6.89 (0.13) to 6.63 (0.18) [group $\times$ time interaction: $F(1,15) = 3.50$, $P = 0.08$, ES: 1.07; ± 1.11].

Mean $\beta_{m_{in\ vitro}}$ did not change for either group following SIT1 [time main effect: $F(1,15) = 0.008$, $P = 0.93$, pooled ES: 0.03; ± 0.65]. $\beta_{m_{in\ vitro}}$ was lower Pre for *Rest-5* than *Rest-1*, 146.7 (11.4) and 134.8 (9.5) mmol H^+ ·kg dry mass $^{-1}$ ·pH $^{-1}$, respectively [group main effect: $F(1,15) = 14.4$, $P = 0.002$, Pre-ES: 1.06; ± 1.37]. There was also no change in mean nonprotein $\beta_{m_{in\ vitro}}$ after SIT1 [time main effect: $F(1,15) = 0.66$, $P = 0.43$, pooled ES: 0.29; ± 0.63] nor was there a meaningful difference between groups [group main effect: $F(1,15) = 1.92$, $P = 0.19$, ES: 0.48; ± 1.36]. The measured nonprotein $\beta_{m_{in\ vitro}}$ at rest was 62.6 (8.0) mmol H^+ ·kg dry mass $^{-1}$ ·pH $^{-1}$, which was 44 (6)% of total $\beta_{m_{in\ vitro}}$ (range: 34–59%). Therefore, the estimated protein $\beta_{m_{in\ vitro}}$ was 76.5 (9.1) mmol H^+ ·kg dry mass $^{-1}$ ·pH $^{-1}$.

Venous blood pH decreased from 7.36 (0.03) Pre to 7.21 (0.07) at +2 min following SIT1 [time main effect: $F(5,68.4) = 34.1$, $P < 0.001$]. There were no clear differences in blood pH between the two groups [group $\times$ time interaction: $F(5,68.4) = 0.48$, $P = 0.79$] because of the high within-group variability (between-group +2 min ES: 0.97; ± 2.13). Mean venous blood $[La^-]$ peaked at +7 min (6.60 and 5.50 mM for *Rest-1* and *Rest-5*, respectively), with a greater increase apparent for *Rest-1* at +5 min and +7 min [group $\times$ time interaction: $F(5,64.7) = 2.77$, $P = 0.025$, +5 min ES: 1.08, $\times / \div 1.12$, +7 min ES: 1.24, $\times / \div 1.16$].

Protein Abundance

MCT1. The MCT1 antibody recognized a single band at ~50 kDa (Fig. 4A). MCT1 protein content did not differ between the two groups [group $\times$ time interaction: $F(2,25.2) = 0.17$, $P = 0.84$, +4 wk ES: 0.04, $\times / \div 0.57$]. On pooling the data there was an increase at +4 wk of 1.18-fold $\times / \div 1.23$ [time main effect: $F(2,25.2) = 3.18$, $P = 0.06$, ES: 0.44, $\times / \div 0.27$]. When the pooled difference scores were compared from +4 wk to +10 wk, i.e., when training volume reduced from 3 to 1

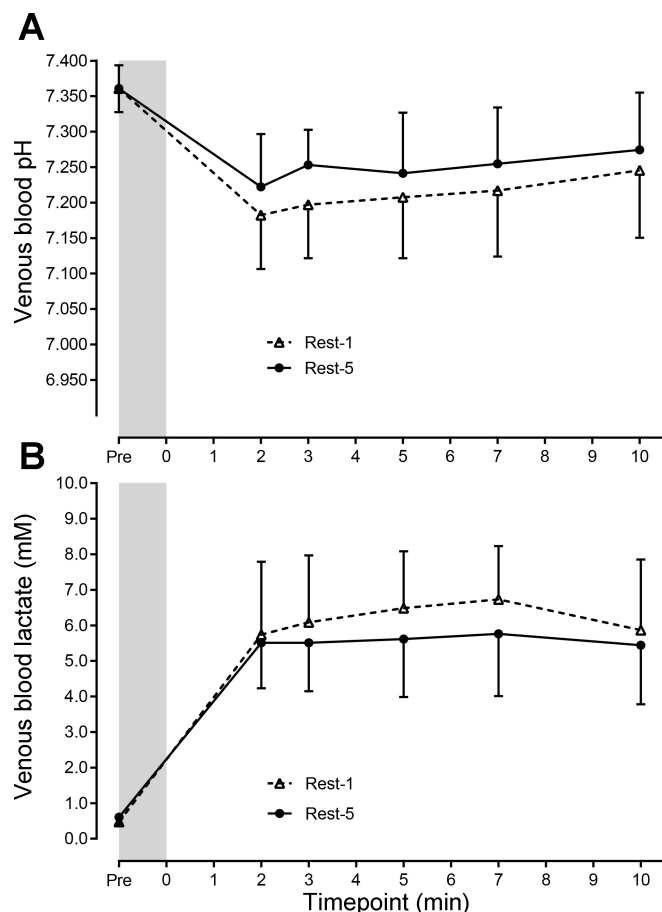


Fig. 3. Venous blood pH (A) and venous lactate (B) before and 2, 3, 5, 7, and 10 min after SIT1 for *Rest-1* (triangles, $n = 7$) and *Rest-5* (circles, $n = 8$); including one participant who dropped out after SIT1). Data are mean (SD). CV of duplicate samples: pH (0.05%), lactate (5.3%).

day/wk, there was no change in mean MCT1 abundance (ES: -0.13 , $\times / \div 0.37$).

MCT4. The MCT4 antibody recognized a strong band at ~ 50 kDa and a weaker band at ~ 75 kDa in some samples (Fig. 4B); only the 50-kDa band was quantified. There were no differences between the two groups [group $\times$ time interaction: $F(2,28.1) = 0.30$, $P = 0.74$, +4 wk ES: 0.21 , $\times / \div 0.82$]. Overall, mean MCT4 abundance did not change in response to training [time main effect: $F(2,28.1) = 0.46$, $P = 0.64$, pooled +4 wk ES: 0.09 , $\times / \div 0.39$].

Basigin. The basigin antibody detected a single band just above 37 kDa (Fig. 4C), likely the canonical basigin-2 isoform (63). Basigin content did not differ between the two groups [group $\times$ time interaction: $F(2,28.3) = 0.20$, $P = 0.82$, +4 wk ES: 0.06 , $\times / \div 0.61$]. On pooling the data there were no meaningful changes [time main effect: $F(2,28.3) = 3.04$, $P = 0.06$, +4 wk ES: 0.30 , $\times / \div 0.29$, +10 wk ES: 0.02 , $\times / \div 0.31$].

NHE1. The NHE1 antibody recognized a single or a double band just below 100 kDa (Fig. 4D). The predicted molecular mass of NHE1 is 91 kDa; therefore, it is probable that the identified band(s) represent non- or partially glycosylated NHE1 protein (39). As per previous reports, both bands were quantified if present (55). Overall there was an increase in NHE1 abundance following training [time main effect: $F(2,25.3) = 4.00$, $P = 0.03$, pooled +4 wk ES: 0.57 , $\times / \div 0.33$]. The two groups responded differently to training [group $\times$ time interaction: $F(2,25.3) = 5.23$, $P = 0.013$]. There were similar increases at +4 wk of 1.11-fold $\times / \div 1.09$ and 1.15-fold $\times / \div 1.22$, for *Rest-1* and *Rest-5*, respectively (ES: 0.16 , $\times / \div 0.80$), but when training volume was reduced to 1 day/wk NHE1 decreased to baseline for *Rest-1*, whereas for *Rest-5* NHE1 increased 1.44-fold $\times / \div 1.18$ compared with baseline (+10 wk ES: 1.68 , $\times / \div 0.97$).

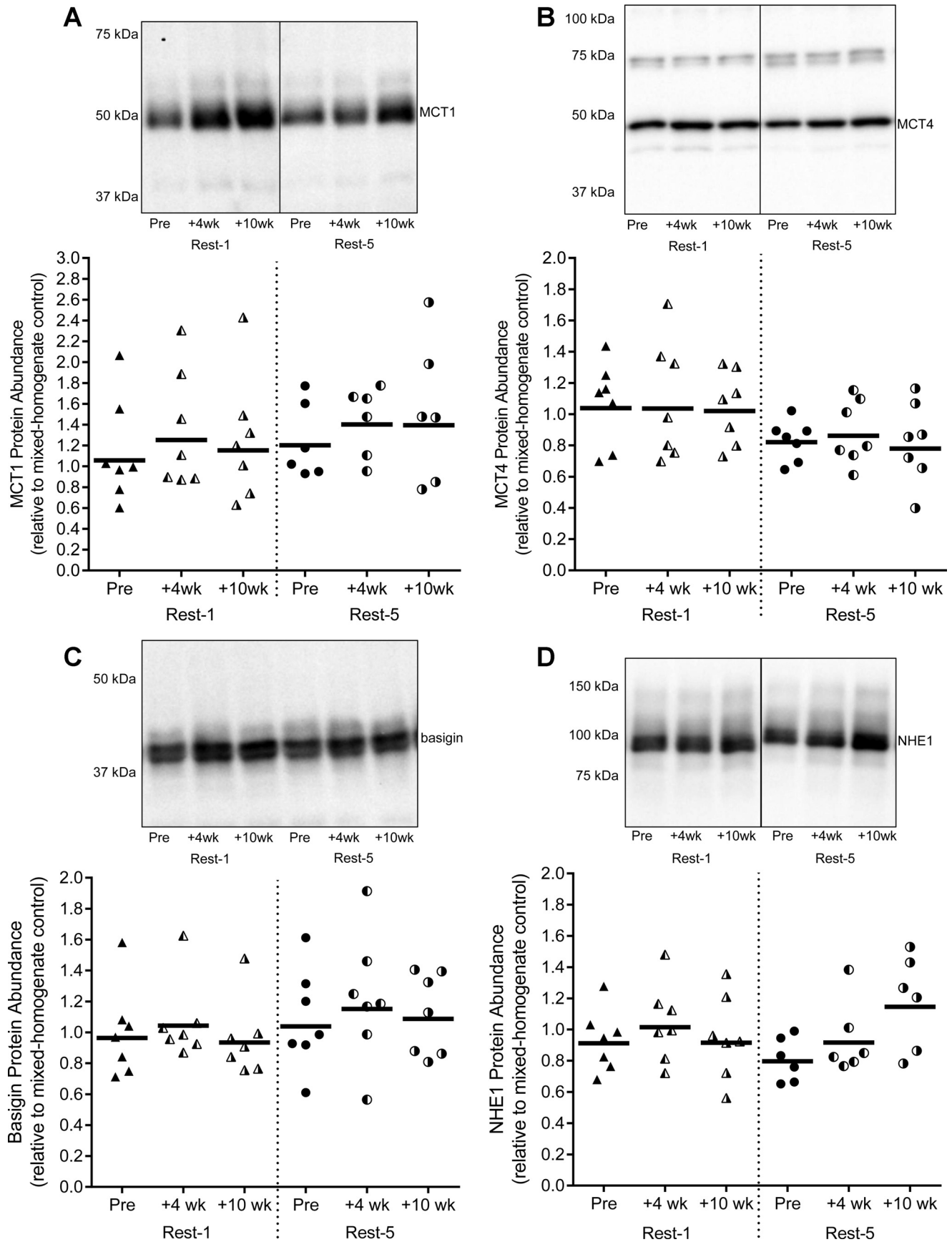
NBCe1. The NBCe1 antibody recognized a band close to the predicted molecular mass of 121 kDa for the canonical NBCe1-B splice variant (Fig. 5). Stronger signal bands of unknown origin were also detected at ~ 50 and 75 kDa; only the ~ 120 -kDa band was quantified (63). Overall there was no meaningful change in NBCe1 abundance following training [time main effect: $F(2,17.8) = 0.23$, $P = 0.80$, pooled +4 wk ES: 0.15 , $\times / \div 0.38$]. Mean NBCe1 content increased for *Rest-1* (1.30-fold $\times / \div 1.55$) and decreased for *Rest-5* (0.89-fold $\times / \div 1.47$) after 4 wk of SIT because of high individual variability/outliers [group $\times$ time interaction: $F(2,17.8) = 1.14$, $P = 0.34$, +4 wk ES: 0.64 , $\times / \div 0.72$]. There was no meaningful difference between groups at +10 wk (ES: 0.42 , $\times / \div 0.94$).

CAII. The CAII antibody recognized a single band just above 25 kDa (Fig. 6A). There was no difference between groups in CAII content over time [group $\times$ time interaction: $F(2,24.9) = 0.08$, $P = 0.92$, +4 wk ES: 0.09 , $\times / \div 0.64$]. On pooling the data, CAII content decreased 0.92-fold $\times / \div 1.11$ following training [time main effect: $F(2,24.9) = 3.79$, $P = 0.04$, +4 wk ES: -0.47 , $\times / \div 0.31$].

CAIII. The CAIII antibody recognized a single band just above 25 kDa (Fig. 6B). There was no overall change in CAIII content for the two groups [time main effect: $F(2,26.9) = 0.32$, $P = 0.73$, pooled +4 wk ES: 0.19 , $\times / \div 0.42$]. There was no difference between the two groups after 4 wk training [group $\times$ time interaction: $F(2,26.9) = 1.74$, $P = 0.20$, ES: 0.26 , $\times / \div 0.91$]. At +10 wk the difference between groups was unclear (ES: 1.04 , $\times / \div 1.42$).

CAIV. The CAIV antibody had poor signal-to-noise ratio but recognized a single or double band near the predicted molecular mass of 35 kDa (Fig. 6C) in the muscle homogenate and positive control (293T Cell Transient Overexpression Lysate; Abnova, Taipei, Taiwan). The CAIV response to training did not differ between the two groups [group $\times$ time interaction: $F(2,18.6) = 0.26$, $P = 0.77$, +4 wk ES: 0.01 , $\times / \div 1.06$]. On pooling the data, CAIV content progressively decreased following training, 0.87-fold $\times / \div 1.40$ and 0.79-fold $\times / \div 1.36$

Fig. 4. Representative immunoblots and protein abundance of monocarboxylate transporter (MCT)-1 (A), MCT4 (B), basigin (C), and sodium/hydrogen exchanger (NHE)-1 (D) before and after 4 (+4 wk) and 10 (+10 wk) wk of sprint interval training (SIT) for *Rest-1* and *Rest-5*. Protein abundance is relative to a calibration curve for a mixed-homogenate internal standard run on every gel. Individual data points and geometric means (horizontal bars) are plotted. Nonadjacent lanes from the same blots are indicated by vertical lines (A, B, and D). CV of duplicate or triplicate gels: MCT1 (16%), MCT4 (16%), basigin (11%), and NHE1 (15%).



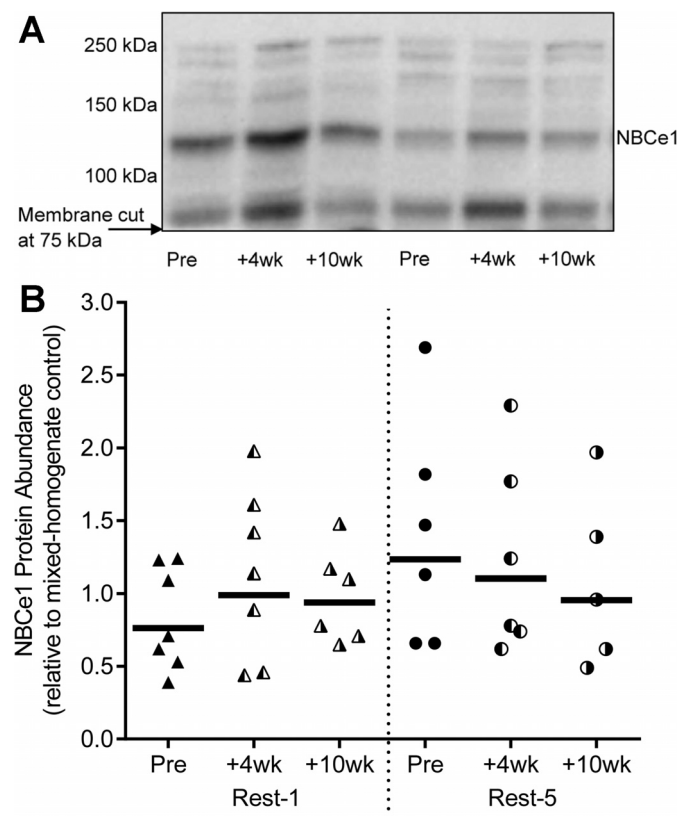


Fig. 5. Representative immunoblot (A) and electrogenic sodium/bicarbonate cotransporter (NBCe1) protein abundance (B) before and after 4 and 10 wk of SIT for *Rest-1* and *Rest-5*. Protein abundance is relative to a calibration curve for a mixed-homogenate internal standard run on every gel. Individual data points and geometric means (horizontal bars) are plotted. CV of duplicate gels was 26%.

at +4 wk and +10 wk, respectively [time main effect: $F(2,18.6) = 3.50$, $P = 0.051$, pooled +4 wk ES: -0.39 , $\times / \div 0.47$, +10 wk ES: -0.72 , $\times / \div 0.46$].

CAXIV. The CAXIV antibody recognized a single band below 50 kDa in the muscle homogenate (predicted molecular mass 38 kDa) and the positive control (Fig. 6D). There was no difference between groups in the CAXIV response [group $\times$ time interaction: $F(2,26.0) = 0.40$, $P = 0.68$, +4 wk ES: 0.08 , $\times / \div 0.96$]. Overall, CAXIV abundance increased in response to training 1.63-fold $\times / \div 1.66$ [time main effect: $F(2,26.0) = 5.77$, $P = 0.008$, pooled +4 wk ES: 1.00 , $\times / \div 0.49$]. After SIT volume was reduced to 1 day/wk, mean CAXIV abundance decreased to baseline values (pooled +10 wk ES: 0.20 , $\times / \div 0.62$).

Muscle Buffer Capacity

$\beta_{\text{in vitro}}$ did not change in response to training for either group (Fig. 7) [time main effect: $F(2,32.0) = 0.11$, $P = 0.89$, pooled +4 wk ES: 0.11 ; ± 0.39]. As noted earlier, $\beta_{\text{in vitro}}$

was lower throughout for *Rest-5* [group main effect: $F(1,15) = 14.4$, $P = 0.002$, Pre-ES: -1.06 ; ± 0.87].

Repeated-Sprint Ability

Total work performed during the RSA test improved only in the *Rest-5* group (Fig. 8A) [group $\times$ time interaction: $F(2,27.0) = 4.60$, $P = 0.02$, +4 wk ES: 0.51 ; ± 0.37]. On reducing SIT volume to 1 day/wk, total work for *Rest-5* decreased slightly from 13.9 (2.2) to 13.5 (2.6) kJ, whereas for *Rest-1* it did not change from 14.0 (2.1) kJ at +4 wk to 14.0 (2.5) kJ at +10 wk. Hence, there was still a clear difference between groups at +10 wk (ES: 0.42 ; ± 0.50). When comparing the change in work performed during each individual sprint (Fig. 8C), *Rest-5* had better improvements for the last three sprints at +4 wk compared with *Rest-1* (sprint 3 ES: 0.57 ; ± 0.49 , sprint 4 ES: 0.78 ; ± 0.50 , sprint 5 ES: 0.51 ; ± 0.46). For both groups work decrement during the RSA test improved following 4 wk of training (Fig. 8B) [time main effect: $F(2,27.4) = 9.57$, $P = 0.001$, pooled +4 wk ES: 0.95 ; ± 0.52]. There were no clear differences between the two groups [group $\times$ time interaction: $F(2,27.4) = 1.84$, $P = 0.18$, +4 wk ES: 0.67 ; ± 1.05 , +10 wk ES: 0.04 ; ± 1.08]. There was no difference between the pooled +4 wk and +10 wk data, indicating that better work decrement was maintained with 1 day/wk training (pooled ES: -0.20 ; ± 0.32).

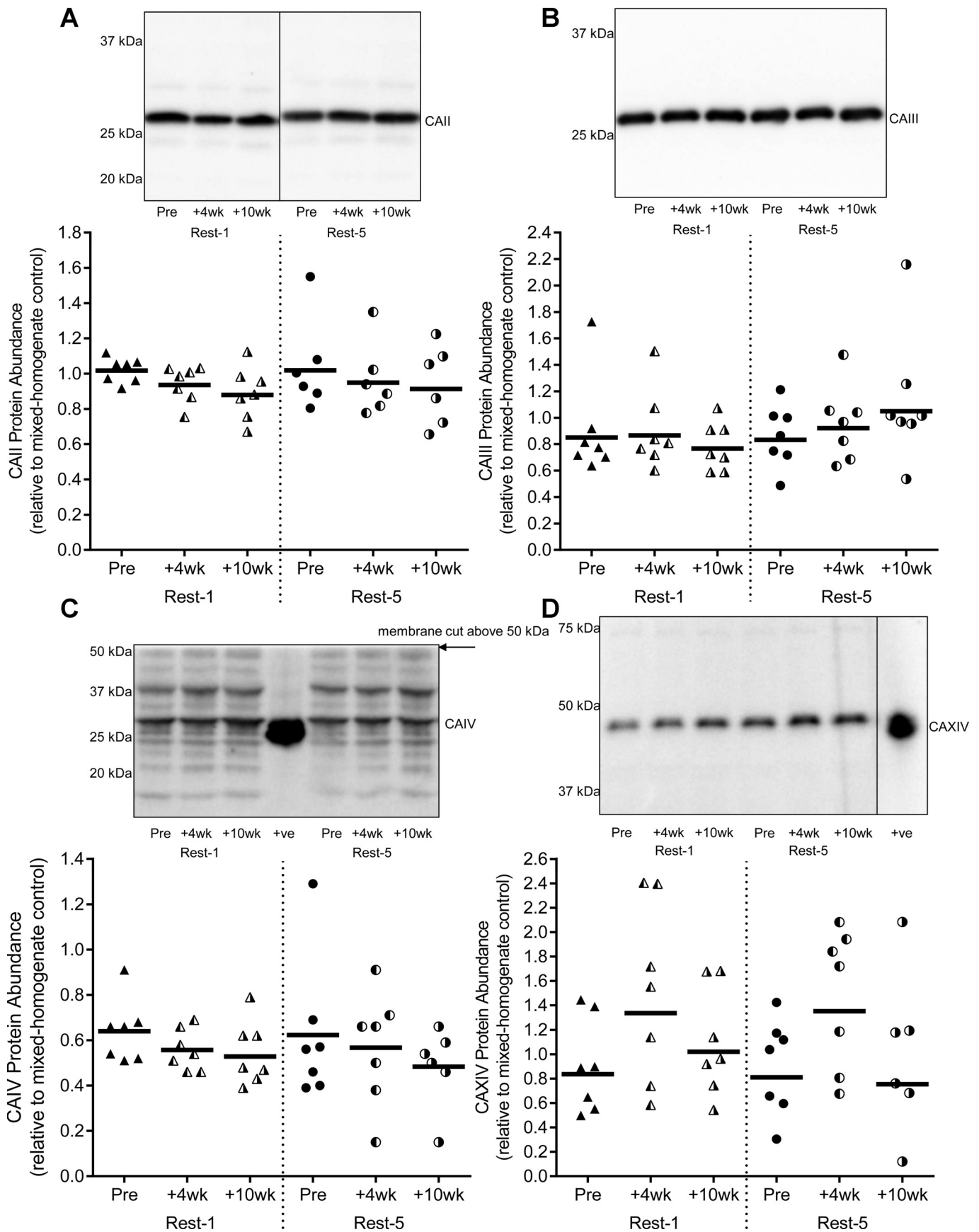
Aerobic Capacity

Power at the LT changed little in response to training for either group [time main effect: $F(1,14) = 2.47$, $P = 0.14$, pooled ES: 0.24 ; ± 0.59] (Fig. 9A). There was also no clear difference between the two groups [group $\times$ time interaction: $F(1,14) = 1.79$, $P = 0.20$, ES: 0.41 ; ± 0.59]. As with the LT, mean $\dot{W}_{\text{peak}}$ did not change with training [time main effect: $F(1,14) = 2.84$, $P = 0.11$, pooled ES: 0.19 ; ± 0.21], nor was there a difference between the two groups [group $\times$ time interaction: $F(1,14) = 0.23$, $P = 0.64$, ES: 0.10 ; ± 0.42] (Fig. 9B). $\dot{V}\text{O}_{2\text{peak}}$ did not change differently between the two groups [group $\times$ time interaction: $F(1,14) = 0.01$, $P = 0.91$, ES: 0.03 ; ± 0.44] (Fig. 9C). On pooling the data there was a small increase in $\dot{V}\text{O}_{2\text{peak}}$ of $2.2 \text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ (90% CL $\pm 1.1 \text{ ml}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$) [time main effect: $F(1,14) = 12.00$, $P = 0.004$, pooled ES: 0.38 ; ± 0.20].

DISCUSSION

This study provides a comprehensive analysis of the acid/base transport protein response to SIT in active women. Our main finding was that selective upregulation of these proteins was evident following SIT, but manipulation of rest interval duration (1 or 5 min) did not influence the adaptive response. Increased abundance of MCT1, NHE1, and CAXIV was seen after 4 wk, with either no change or a decrease evident for MCT4, NBCe1, basigin, CAII, CAIII, and CAIV. There was also no change in $\beta_{\text{in vitro}}$. Greater improvements in RSA

Fig. 6. Representative immunoblots and protein abundance of carbonic anhydrase (CA) II (A), CAIII (B), CAIV (C), and CAXIV (D) before and after 4 and 10 wk of SIT for *Rest-1* and *Rest-5*. Protein abundance is relative to a calibration curve for a mixed-homogenate internal standard run on every gel. Individual data points and geometric means (horizontal bars) are plotted. Positive controls (+ve) were loaded for CAIV and CAXIV. Nonadjacent lanes from the same blots are indicated by vertical lines (A and D). CV of duplicate or triplicate gels: CAII (15%), CAIII (single gels), CAIV (23%), and CAXIV (29%).



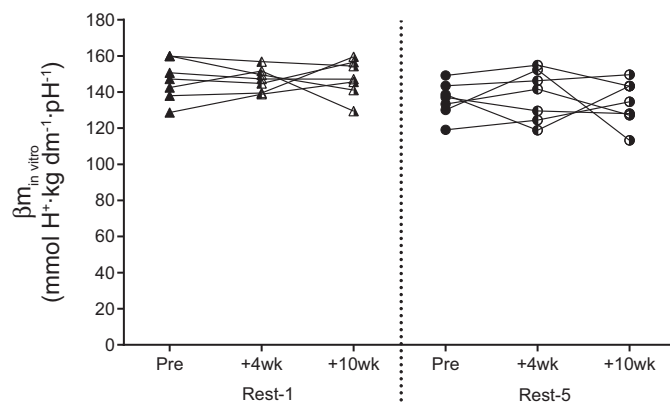


Fig. 7. $\beta_{m \text{ in vitro}}$ before and after 4 and 10 wk of SIT for *Rest-1* and *Rest-5*. Individual data points are plotted. CV of duplicate titrations = 7.7%.

were seen with *Rest-5* compared with *Rest-1*, whereas both groups had similar improvements in aerobic capacity. A second major finding was that enhanced protein abundance and RSA were maintained when training volume was reduced to 1 day/wk, although disparate responses between groups were seen for NHE1. Overall, these data demonstrate that short-term supramaximal-intensity exercise has modest effects on pH-regulatory components compared with, for example, the rapid mitochondrial remodeling reported following SIT (30).

MCT1 protein content increased for both *Rest-1* and *Rest-5*, but in common with most studies there were no changes in MCT4 protein content following 4 wk of SIT (3, 10, 32, 49, 66, 76). Although the signalling factors responsible for regulation of the MCTs in vivo remain to be determined, the current study supports the existing in vitro evidence that they respond to distinct stimuli (34, 38, 79). For example, increased content of MCT1, but not MCT4, has been shown in L6 cells 1 h after incubation with either 10 or 20 mM La^- (38). In our study, immediately after SIT1, muscle $[\text{La}^-]$ increased to 25.5 (6.2) and 18.7 (9.7) mM for *Rest-1* and *Rest-5*, respectively. Therefore, consistent with in vitro data (38), intracellular La^- accumulation may be a stimulus for MCT1, but not MCT4, upregulation in human skeletal muscle. However, given the similar increase in MCT1 abundance for both groups, and that exercise-induced decreases in pH_i differed by only ~ 0.1 pH unit following SIT1, our data prove inconclusive as to whether H^+ accumulation is an important signal for inducing MCT1 expression. The absence of any change in MCT4 abundance indicates that neither La^- nor H^+ accumulation is likely to be an important signalling factor in MCT4 regulation.

The factors affecting regulation of the MCT chaperone protein basigin in human skeletal muscle have not been described. The present study provides the first investigation of the response of basigin to a SIT intervention, finding no meaningful changes in protein abundance after 4 or 10 wk of training for either group. Given the substantial increase in muscle $[\text{La}^-]$ following both *Rest-1* and *Rest-5* noted above, these results contrast with previous data from L6 cells (38), whereby basigin protein content increased by 1.85- to 2.78-fold after 1 h of La^- incubation (10 or 20 mM) but decreased after 6–48 h of incubation. However, a difference between the cell model and exercising human muscle is that prolonged $[\text{La}^-]$ of 10 or 20 mM are not physiological in human muscle in vivo. Whereas

accumulation of similar levels of intracellular La^- was found in the present study, and is typical of high-intensity exercise, activity of the MCTs causes La^- efflux from the cytosol, resulting in oxidation of La^- to pyruvate or entry of La^- in the Cori cycle for gluconeogenesis (24). Thus, although prolonged high $[\text{La}^-]$ may stimulate an increase in basigin content in L6 cells, the present data suggest the high $[\text{La}^-]$ typical of SIT does not provoke increases in basigin content in exercising human muscle.

Second to the MCTs, NHE1 provides the next most important contribution to H^+ efflux during high-intensity exercise (4, 5). This study found small increases in NHE1 abundance after 4 wk of SIT for *Rest-1* (1.12-fold $\times / \div 1.09$) and *Rest-5* (1.13-fold $\times / \div 1.22$). All of the research to date on the NHE1 protein response to training has emanated from Copenhagen (3, 32, 33, 49, 54, 66, 78, 81), with mean fold-change following HIIT/SIT interventions ranging from 0.95-fold (81) to 1.35-fold (78). The one study to include females, in a group consisting of both sexes, found a 1.14-fold increase with training (32). Thus, it seems, while NHE1 abundance may increase following HIIT/SIT in different populations, the magnitude of any increase is typically small. An unexpected finding in the current study was a further increase in NHE1 content for *Rest-5* of 1.25-fold $\times / \div 1.23$, compared with +4 wk, after SIT volume was reduced to 1 day/wk. In contrast, NHE1 abundance returned to baseline for *Rest-1*, indicating the stress imposed by the two training regimens differentially induced NHE1, but only when training load was reduced. One study has shown NHE1 abundance to increase 1.35-fold in elite footballers who stopped training for 2 wk at the end of their season, in contrast to a group who performed 2 wk of mixed HIIT/SIT (81). It appears that, following a period of intensive training, a reduced training stimulus is favorable for NHE1 upregulation. This may relate to the long fractional degradation rate for NHE1 mRNA whereby steady-state mRNA content is not apparent until 24–48 h post-HIIT (62). If there is a subsequent delay in protein synthesis, then upregulation of NHE1 may benefit from a longer duration between exercise sessions. Further evidence is required to confirm this.

The NBCe1 protein response to a SIT intervention has been reported here for the first time. Although no changes in mean NBCe1 content were seen for either group, there were highly variable responses. We previously found NBCe1 abundance to increase in men undergoing 4 wk of HIIT performed at different intensities between the LT and W_{peak} (63). Using a non-isoform-specific antibody, others have reported NBC content to increase in rat soleus following 5 wk of HIIT, regardless of whether rats received sodium bicarbonate or a placebo before each training session (80). By buffering extracellular H^+ , sodium bicarbonate enhances H^+ efflux by creating a greater pH gradient relative to the intracellular space (58). In the current study, 5 min of rest would also have enabled greater H^+ efflux compared with 1 min of rest. Therefore, from these limited data, it seems that H^+ efflux does not influence changes in NBC(e1) protein content. Although there may be sex-specific responses on comparing our two human studies, it seems that longer-duration intervals of high-intensity training may be required to provoke adaptations in NBCe1. The signaling factors required for regulation of NBCe1 in human muscle in vivo are unknown, but in vitro research has shown that inositol 1,4,5-trisphosphate receptor-binding protein re-

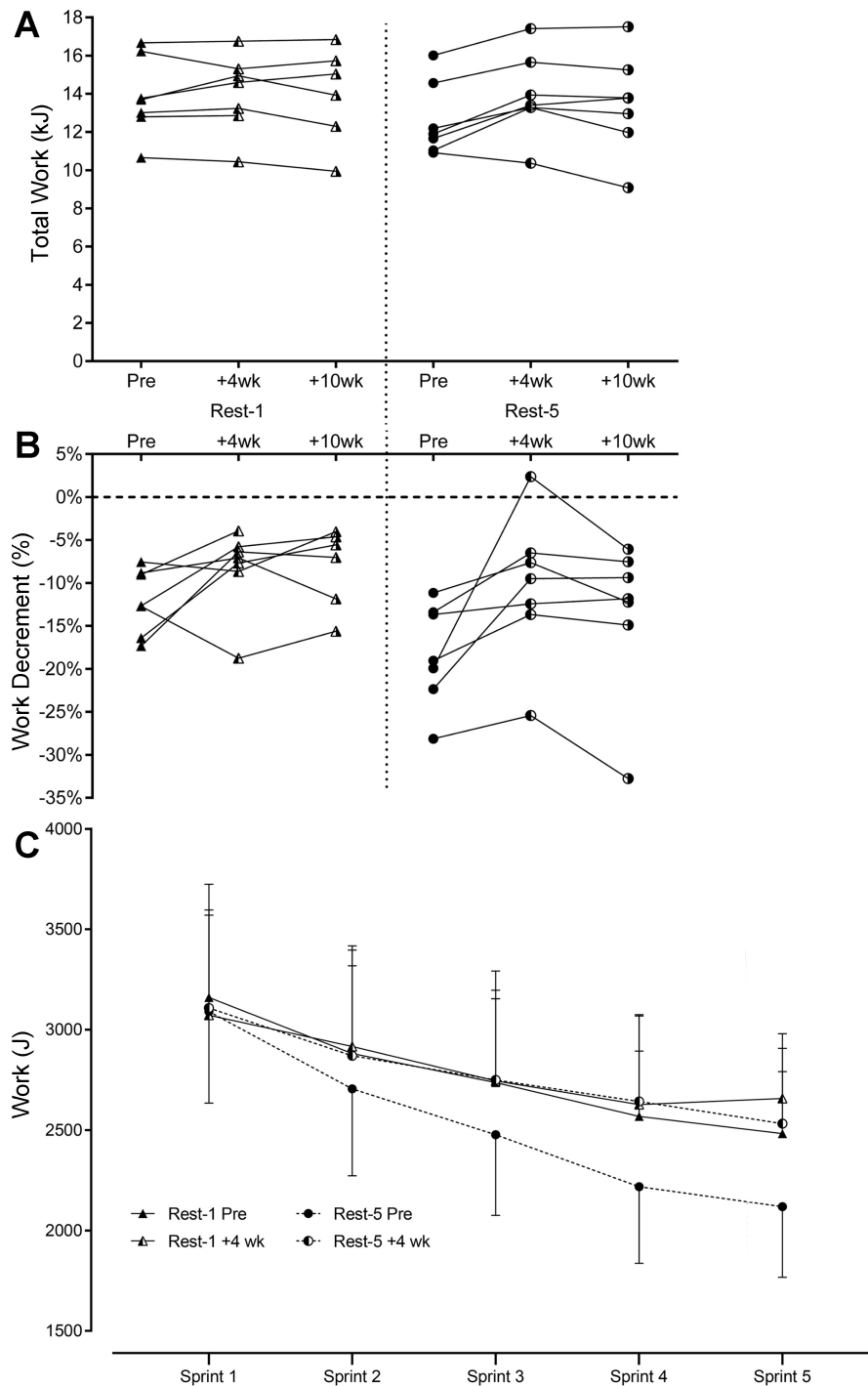


Fig. 8. Total work (A) and work decrement during a repeated-sprint ability test (5×6 s) (B) before and after 4 and 10 wk of SIT for *Rest-1* and *Rest-5* (individual data points) and work performed per individual sprint for *Rest-1* (triangles) and *Rest-5* (circles) before and after 4 wk of SIT [mean (SD)] (C).

leased with inositol 1,4,5-trisphosphate (IRBIT) activates most of the NCBT proteins, except for NBCe2 (73). Whether IRBIT is important in vivo remains to be shown, but it is a potential target for future study.

There were contrasting changes in each of the CA isozymes after 4 wk of training. Of the cytosolic CAs, CAII content decreased slightly for both groups, whereas there was no change in CAIII content. Of the sarcolemmal CAs, there were large increases in CAXIV abundance for both *Rest-1* and *Rest-5* but a small decrease in CAIV content for both groups at +4 wk. There are no published data on the CA response to SIT,

although we reported increases in CAII and CAXIV content, but decreases or no change in CAIII or CAIV, following 4 wk of 2 min HIIT in men (63). Given there were no differences between groups in the present study, despite differing metabolic stresses imposed by 1- vs. 5-min rest intervals, nor after differing intensities of HIIT (63), there is little insight provided as to the potential mechanisms for regulation of CA content.

There were no changes in $\beta_{\text{min vitro}}$ for either group following 4 or 10 wk of SIT. Previous research postulated that H^+ accumulation might be a stimulus for upregulation of $\beta_{\text{min vitro}}$ (21, 83). If so, then $\beta_{\text{min vitro}}$ might have been expected to

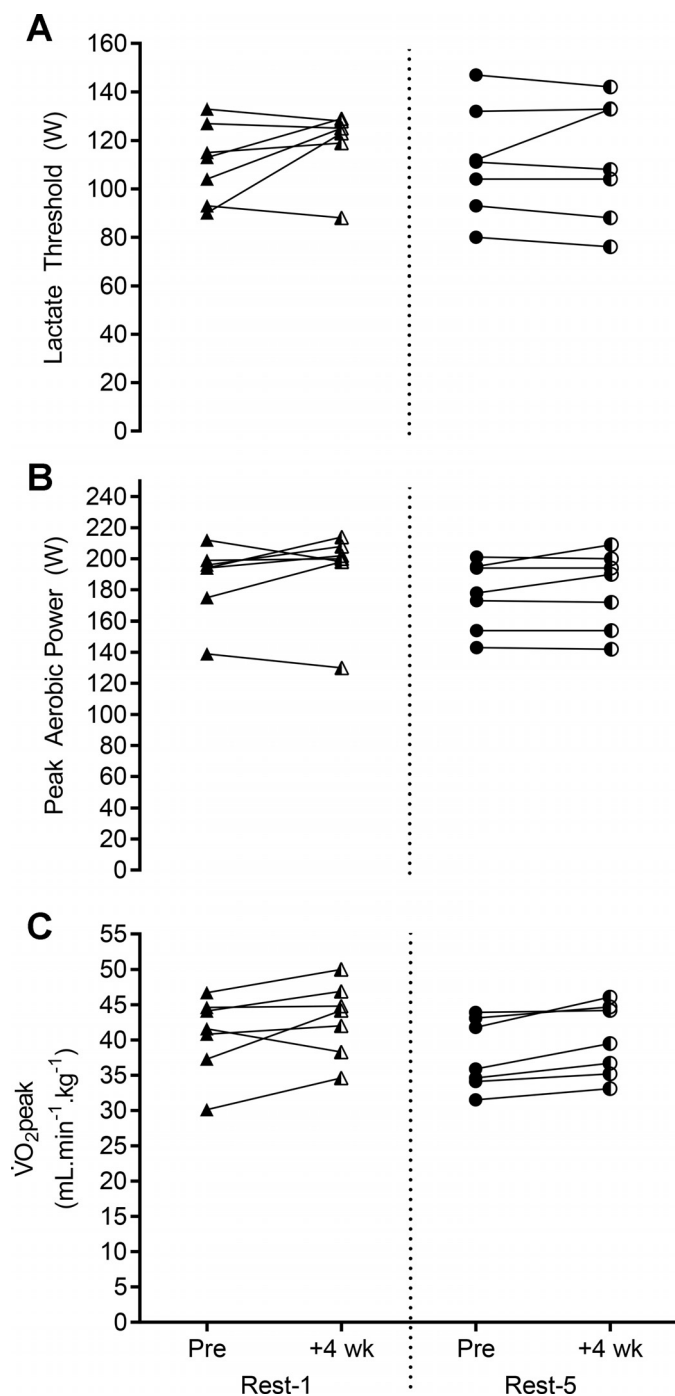


Fig. 9. Lactate threshold (A), peak aerobic power (W_{peak} , B), and peak oxygen uptake ($\dot{V}O_{2peak}$, C) before and after 4 wk of SIT for *Rest-1* and *Rest-5*. Individual data points are plotted.

improve more for *Rest-1* because the shorter rest intervals would prevent recovery of pH_i between sprints. As expected, muscle pH was lower for *Rest-1* than *Rest-5*, but, given there were no improvements in $\beta_{m_{in\ vitro}}$ for either group, the low muscle pH associated with both types of training was not a factor in improving $\beta_{m_{in\ vitro}}$. These data add to the existing evidence showing improvements in $\beta_{m_{in\ vitro}}$ are typically not found with short-duration work intervals (2, 20, 37, 49, 71).

To profile the metabolic stress imposed by the two exercise bouts, we also assayed total and nonprotein $\beta_{m_{in\ vitro}}$ before

and after the first bout of SIT. Although a small reduction in $\beta_{m_{in\ vitro}}$ might theoretically be expected postexercise because of the greater content of organic phosphates and lower free P_i content, mean nonprotein $\beta_{m_{in\ vitro}}$ changed little. The randomness of individual responses suggests methodological rather than physiological variation. Our group has previously reported lower $\beta_{m_{in\ vitro}}$ immediately after various bouts of high-intensity exercise (11, 13). In one study, because nonprotein $\beta_{m_{in\ vitro}}$ was unchanged, the lower total $\beta_{m_{in\ vitro}}$ was inferred to be the result of reduced protein buffering (11). However, in the current study, with no change in either total $\beta_{m_{in\ vitro}}$ or nonprotein $\beta_{m_{in\ vitro}}$, there was consequently no change in estimated protein $\beta_{m_{in\ vitro}}$. Therefore, the present data cast doubt on whether a short-term reduction in protein buffering is likely immediately following high-intensity exercise. This adds to our data questioning the sensitivity and ecological validity of the titration assay (62). Notwithstanding these concerns, our estimated protein $\beta_{m_{in\ vitro}}$ of 76.5 (9.1) mmol H^+ ·kg dry mass⁻¹·pH⁻¹, or 23.2 (2.8) mmol·l⁻¹·pH⁻¹, is similar to the 26 mmol·l⁻¹·pH⁻¹ predicted from the Henderson-Hasselbalch equation for protein-bound histidine (48).

We anticipated that performance improvements would be specific to training that more closely matched the substrate usage of the exercise test; however, not all of the performance improvements were as expected. RSA did improve more for *Rest-5*, with greater total work performed and better fatigue resistance. This was not the result of improved single (i.e., first) sprint performance but because of progressively greater total work performed in sequential sprints at +4 wk compared with Pre. Greater total work probably relates to an enhanced anaerobic capacity providing greater relative substrate level phosphorylation for sequential sprints, in part because of improved PCr resynthesis (12). *Rest-1* had no improvement in total work but did have improved fatigue resistance because of a lower reduction in work for the later sprints. Better fatigue resistance more likely reflects that the ability to sustain power for later sprints is dependent on oxidative rather than glycolytic capacity (61). Accordingly, both groups displayed similarly modest improvements in $\dot{V}O_{2peak}$ of ~6%, or 2.2 (90% CL $\pm$ 2.4 mL·min⁻¹·kg⁻¹) and 2.1 $\pm$ 1.0 mL·min⁻¹·kg⁻¹ for *Rest-1* and *Rest-5*, respectively. These data are less than the aggregate reported in a meta-analysis of SIT studies of 8%, or 3.6 mL·min⁻¹·kg⁻¹ (31), but greater than the 3.6% improvement in a subgroup analysis of nonathletic active females, reported in a separate meta-analysis (85). It appears that short-duration SIT interventions produce only moderate improvements in $\dot{V}O_{2peak}$ in active women. Finally, there were no group mean improvements in the LT or W_{peak} for *Rest-1* or *Rest-5*, but there was notable individual variability within both groups. Those factors determining the individual training response require further research (59), with particular emphasis on study design to enable robust inferences (1, 40).

We have shown that adaptations in some acid/base transport proteins can be maintained by a reduction in training volume for a subsequent 5-wk period of similar-intensity SIT. One unexpected finding was that NHE1 protein content increased further for *Rest-5* after reducing SIT volume to 1 day/wk. Why this occurred is uncertain, but it may be that with comparatively long mRNA and protein half-lives (62), NHE1 protein synthesis peaks with longer recovery between training sessions. Finally, improved performance was evident during the

RSA test for both groups, but to different degrees. On comparing total work and work decrement at +10 and +4 wk there were no differences, demonstrating that, where RSA was enhanced after 4 wk of training, it was maintained with just 1 day/wk of training at a similar intensity. This confirms that, once performance adaptations are achieved with a short period of intensified training, they can be readily maintained with a lower volume of high-intensity training (41, 42, 46, 47, 67).

Perspectives and Significance

Selective upregulation of acid/base transport proteins was found in active women after 4 wk of SIT. The similar adaptations of two work-matched training groups, differing only in rest intervals of 1 or 5 min, indicate duration of recovery does not influence short-term adaptations in these proteins, at least not in an active female population. We also report evidence supporting lactate as a signaling molecule for inducing MCT1, but not MCT4, expression. Furthermore, 4 wk of SIT produced only modest improvements in $\dot{V}O_{2\text{peak}}$, indicating there was insufficient training volume to produce the primarily central adaptations that determine $\dot{V}O_{2\text{peak}}$. Overall, there was little change in mean LT or W_{peak} , but possible evidence of high and low responders in both groups merits further exploration. Meanwhile, there were better improvements in RSA for Rest-5, likely reflecting the greater anaerobic training stimulus provided by longer rest intervals. Finally, maintaining the physiological stimulus of training intensity, but reducing the volume of training, was sufficient to mitigate reversal of adaptations for acid/base transport proteins and RSA. The implication from this research is that rest intervals of 5 min during SIT lead to better improvements in RSA than 1-min rest intervals, and these improvements can be maintained despite a reduction in the volume of SIT performed.

ACKNOWLEDGMENTS

We are very grateful to the participants for their effort and commitment to this study. We also thank Dr. Mitch Anderson and Dr. Andrew Garnham for performing the muscle biopsies.

GRANTS

This research was supported in part by an Applied Sports Science Research Grant from Exercise & Sports Science Australia.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

C.M. and D.J.B. conceived and designed research; C.M. performed experiments; C.M. analyzed data; C.M. and D.J.B. interpreted results of experiments; C.M. prepared figures; C.M. drafted manuscript; C.M. and D.J.B. edited and revised manuscript; C.M. and D.J.B. approved final version of manuscript.

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